

ZEN Univ.

IUT 1-2

Exercise



1. Assume there are finite no. of $4n+3$ primes.

Let them be $p_1, \dots, p_k$ ($k \in \mathbb{N}$)

Consider $N := 2 \cdot p_1 \cdot \dots \cdot p_k + 1$

$$N \equiv 2 \cdot 3^k + 1 \pmod{4}$$

By checking mod 2 of k , we can get $N \equiv 3 \pmod{4}$. (*)

N is not dividable by p_i (or 2) by checking mod p_i (or 2)

So, N must only have $(4n+1)$ -formed prime factors,

but they are 1 in mod 4, this contradicts with (*). $\blacksquare$

2. (a) $a > b > 0, n > 1, a^n - b^n$ is prime

for $a-b=1$)

$$a^n - b^n = (a-b)(a^{n-1} + a^{n-2}b + \dots + ab^{n-2} + b^{n-1})$$

$$\hookrightarrow \left(\sum_{i=0}^{n-1} a^i b^{n-1-i} \right)$$

by $a > b > 0$, it's $\geq a+b \geq 3$
and $n-1 \geq 1$

It's prime, so $(a-b)$ must be 1. N
o

for n) Assume n is not prime. $n = mk$ for some $m, k > 1$

$$a^n - b^n = a^{mk} - b^{mk} \quad \text{Let } A := a^m, B := b^m.$$

$$a^n - b^n = A^k - B^k = (A-B)(A^{k-1} + A^{k-2}B + \dots + B^{k-1}).$$

$\hookrightarrow$

$$\begin{aligned}
 A - B &= a^m - b^m > a^2 b^{m-2} - b^m = b^{m-2} (a^2 - b^2) \\
 &= b^{m-2} (a-b)(a+b) \\
 &\geq a+b \\
 &= 2+1 = 3.
 \end{aligned}$$

by $a > b$
 $m \geq 2$

$$A^{k-1} + A^{k-2}B + \dots + B^{k-1} \geq A^{k-1} + B^{k-1} \geq 2$$

So, at least 2 prime factors included.

This contradicts with $(a-b)$ is prime.

Thus (*) is false. $\odot$

(b) Let's define $d(\cdot)$ as divisor sum function.

Statement can be written as $d(n) = 2n \iff \begin{matrix} 2^m \\ n = 2^{m-1}(2^m - 1) \\ \text{and } (2^m - 1) \text{ is prime.} \end{matrix}$

$$\begin{aligned} (\Leftarrow) \quad d(n) &= d(2^{m-1}) d(2^m - 1) \\ &= (2^m - 1) 2^m = 2n \end{aligned}$$

$$\begin{aligned} d(ab) &= d(a)d(b) \\ \text{if } \gcd(a,b) &= 1 \\ \text{and } 2^{m-1} &\text{ is prime} \end{aligned}$$

$(\Rightarrow)$ Let $n = 2^k s$ for some unique k, s st. $2 \nmid s$.
 n is even, so $k \geq 1$.

$$d(n) = 2n$$

$$d(2^k) d(s) = 2^{k+1} s$$

$$(2^{k+1} - 1) d(s) = 2^{k+1} s$$

$\hookrightarrow$

2^{k+1} and $2^{k+1} - 1$ are mutually prime, so $(2^{k+1} - 1) \mid s$.

Let $s = (2^{k+1} - 1)t$ for some $t \geq 1$

$$(2^{k+1} - 1) d(s) = 2^{k+1} (2^{k+1} - 1) t$$

$$\underline{d((2^{k+1} - 1)t) = 2^{k+1} t} \quad (*)$$

if $t > 1$, s has at least $1, 2^{k+1} - 1, t, (2^{k+1} - 1)t$ as factors.

$$d(s) \geq 1 + (2^{k+1} - 1) + t + (2^{k+1} - 1)t$$

$$= 2^{k+1} + 2^{k+1} t > 2^{k+1} t. \text{ This contradicts with } (*)$$

so t must be 1.

$d(2^{k+1} - 1) = 2^{k+1}$ means $2^{k+1} - 1$ is prime. $n = 2^k (2^{k+1} - 1)$

3. (a) $IJ \subseteq I \cap J$.

Prf. $\forall i \in I, j \in J, ij \in I \cap J$. by I, J is ideal.

IJ is minimal ideal by its definition.

and $I \cap J$ is ideal. so it's upper bound of IJ .

(b) $(I+J)(I \cap J) \subseteq IJ$.

Prf. Generator of $(I+J)(I \cap J)$ forms $(i+j)x$
 $(i \in I, j \in J, x \in I \cap J)$

$$(i+j)x = \overset{I}{\underbrace{i}} \overset{J}{\underbrace{x}} + \overset{I}{\underbrace{x}} \overset{J}{\underbrace{j}}$$

$ix, jx \in IJ$, so all generators is included in IJ .

$$4. \quad (2, X)(3, X) = (6, X).$$

Prf $\Rightarrow$ For all generators has form of (for some $a, b, c, d \in \mathbb{Z}[X]$)

$$(a \cdot 2 + bX)(c \cdot 3 + dX)$$

$$= (ac) \cdot 6 + (2ad + 3cb + bdX)X \\ \in (6, X)$$

$\Leftarrow$ Let's prove $6, X \in (2, X)(3, X)$

$$\begin{array}{rcccl} & (2, X) & (3, X) & & \\ & \downarrow & \downarrow & & \\ 6 & = & 2 \cdot 3 & & \\ & (2, X) & (3, X) & (2, X) & (3, X) \\ & \downarrow & \downarrow & \downarrow & \downarrow \\ X & = & X \cdot 3 & - & 2 \cdot X \end{array}$$

D

$$\underline{X \notin \{ab : a \in (2, x), b \in (3, x)\}} \quad \text{!!} \quad A$$

Assume $X \in A$. $\exists a, b$ $ab = X$. note that $a, b \neq 0$.

$$\deg(a) + \deg(b) = \deg(X) = 1, \text{ so}$$

$$\deg(a) = 1, \deg(b) = 0 \quad (\text{or vice versa}) \quad (*)$$

$$(cX + 2d)(3e) \quad \text{for some } c, d, e \in \mathbb{Z}$$

$$= 6de + 3ecX$$

This can't be X . Similar for $(*)$ 

5.

$$I = \left\{ \sum_{m=1}^n r_m s_m : \underline{r_m \in R, s_m \in S, n \geq 1} \right\} \quad \text{"X"}$$

Pf

$$I \subseteq X)$$

$S \subseteq X$ and X is ideal,

so X is upper bound of I .

$$I \supseteq X) \quad \text{Let } X_N := \left\{ \sum_{m=1}^n r_m s_m : \dots, n \leq N \right\}$$

$$\text{Here, } X = \bigcup_{i \in \mathbb{N}^+} X_i$$

By induction for N , we can prove $\forall N, I \supseteq X_N$

if $N=1$) Obvious.

$$\left(\begin{array}{l} X_1 = \{ rs : r \in R, s \in S \} \\ \text{and } S \subseteq I, \text{ and } I \text{ is ideal, so } rs \in I \end{array} \right)$$

if $N = K+1$, $I \supseteq X_K$)

For all elem in X_K , it forms $\sum_{n=1}^{K+1} r_n s_n = \underbrace{\left(\sum_{n=1}^K r_n s_n \right)}_{\substack{\cap \\ X_K \\ \cap I}} + \underbrace{r_{K+1} s_{K+1}}_{\substack{\cap \\ X_K \\ \cap I}}$

I must be ideal, so addition of them is also included in I $\square$

by
induction
premise